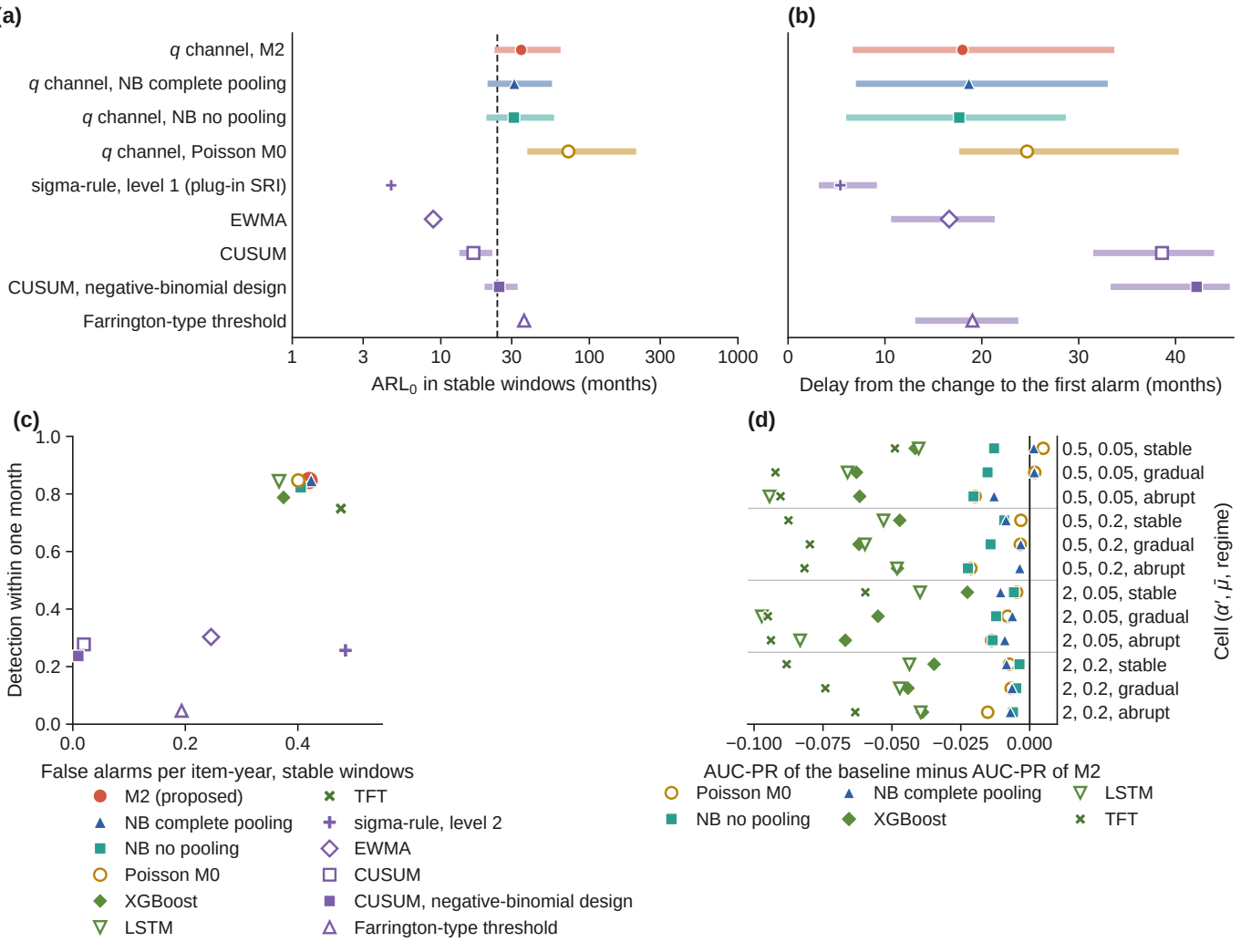
(a)
q channel, M2
q channel, NB complete pooling
q channel, NB no pooling
q channel, Poisson M0
sigma-rule, level 1 (plug-in SRI)
EWMA
CUSUM
CUSUM, negative-binomial design
Farrington-type threshold
1
3
10
30
100
300
1000
ARL0 in stable windows (months)
(b)
0
10
20
30
40
Delay from the change to the first alarm (months)
(c)
Detection within one month
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
False alarms per item-year, stable windows
M2 (proposed)
NB complete pooling
NB no pooling
Poisson M0
XGBoost
LSTM
TFT
sigma-rule, level 2
EWMA
CUSUM
CUSUM, negative-binomial design
Farrington-type threshold
(d)
0.5, 0.05, stable
0.5, 0.05, gradual
0.5, 0.05, abrupt
0.5, 0.2, stable
0.5, 0.2, gradual
0.5, 0.2, abrupt
2, 0.05, stable
2, 0.05, gradual
2, 0.05, abrupt
2, 0.2, stable
2, 0.2, gradual
2, 0.2, abrupt
Cell (α′, μ̄1, regime)
−0.100
−0.075
−0.050
−0.025
0.000
AUC-PR of the baseline minus AUC-PR of M2
Poisson M0
NB complete pooling
LSTM
NB no pooling
XGBoost
TFT